Dhritabrata Mitra
dhritabratamitra2@gmail.com | (810) 531-3212 | Tempe, Arizona | LinkedIn | GitHub | Portfolio

EDUCATION

Research Assistant, Michigan Technological University — Jun 2025 – Present

M.S. in Computer Science, Michigan Technological University, GPA: 3.85/4.0 — Aug 2023 – Apr 2025

B.Tech in Computer Science, University of Engineering & Management, GPA: 3.39/4.0 — Jul 2016 – May 2020

WORK EXPERIENCE

Cognizant — Jan 2020 – Jun 2023

Client: Netflix

Software Engineer II — Jan 2021 – Jun 2023

- Built an **Entitlement & Regional Catalog microservices** using Spring Boot **REST APIs** on AWS; serving 90M+ users with P95 ≤150ms latency and 99.95% uptime, growing a $30B+ streaming platform.
- Saved **$200K/year** by optimizing content search through re-engineering **Elasticsearch** queries, adding **Redis** caching, and tuning **PostgreSQL indexes**, cutting search latency by 40%.
- Diagnosed delayed and duplicate notifications in an **event-driven Kafka pipeline**, implementing idempotent producers, exponential backoff retries, and consumer deduplication, reducing latency by 45% and duplicate incidents by 95%.
- Reduced manual deployment errors by 99% by **building Jenkins CI/CD pipelines**, integrating with AWS CodeDeploy, configuring IAM, and adding AWS Lambda notifications; automated build, test, and release validation.
- **Containerized** React/Next.js frontend and Java services with **Docker** and orchestrated deployments on AWS EKS using **Helm** charts and **ArgoCD (GitOps)**, allowing blue/green, rolling, and canary releases, cutting release cycles by 50%.
- Implemented IaC governance for **Kubernetes** and **Terraform** (policy checks and gated CI), blocking insecure configs before deployment and reducing security/ops exceptions by 40% while improving deployment success to 99%.
- **Mentored 5 interns**, led code reviews, and introduced shared coding standards, reducing post-release defects across multiple deployments by 45%.

Software Engineer I — Jan 2020 – Dec 2020

- Resolved frontend and backend production issues through root cause analysis, reducing downtime from **12 hours per quarter to 30 minutes** and cutting average production incidents from **250 to 20 per month**.
- Eliminated over-fetching, inconsistent pagination, and fragmented responses by optimizing **REST** and **GraphQL API** integrations and standardizing API responses, reducing page errors by 30% and improving responsiveness by 60%.
- Built **CloudWatch** and **Splunk** dashboards and alerts for **Java** and **Kafka** services, and drove runbooks and RCA follow-ups, cutting MTTD by 40%, MTTR by 25%, and recurring defects by 20% during Sev-2/Sev-3 incidents.
- Built and maintained unit, integration, and end-to-end test suites using **JUnit, Mockito, Jest,** and **Playwright**, following Test-Driven Development (TDD) practices, improving backend reliability by 50% and production issues by 70%.

SKILLS

Languages: Java, SQL, Go / Golang, Python, TypeScript, JavaScript

Backend: Spring Boot, Spring MVC, Spring Data JPA, Spring Security, REST APIs, GraphQL APIs, Node.js

SQL/NoSQL Databases: PostgreSQL, Cassandra, Redis, Elasticsearch, MongoDB

Observability & Event Streaming: AWS CloudWatch, Apache Splunk, Apache Kafka, RabbitMQ

Cloud/DevOps: AWS (EC2, EKS, S3, Lambda, CodeDeploy, IAM, CloudWatch), Docker, Kubernetes, ArgoCD (GitOps), Terraform, Helm, Jenkins, Maven, Git, Linux

Testing: JUnit, Mockito, Jest, Playwright

Frontend: Next.js, React.js, Redux, Zustand, Tailwind CSS, HTML5, CSS3

Realtime Communication: WebRTC, WebSockets, Socket.IO

TECHNICAL PROJECTS

BingeWatchIt | GitHub

- A scalable, distributed microservices movie streaming platform, enabling users to upload, stream, and share videos.
- Built with Go, Next.js, TypeScript, Kafka, MongoDB, PostgreSQL, and Tailwind CSS.

Meet AI | Demo | Live Site | GitHub

- Built a SaaS platform with OpenAI LLM integration for real-time verbal Q&A, reducing time spent searching for answers.
- Built with Next.js, Nginx, Tailwind CSS, WebRTC, tRPC, Drizzle ORM, NeonDB, and subscription billing via Polar API.

ChatAway - Real-time Chat App | Demo | Live Site | GitHub

- A real-time chat application, addressing delays in teamwork by providing immediate messaging and media sharing.
- Built with React.js, Tailwind CSS, Socket.IO, OAuth (GCP), JWT auth, and global state management with Zustand.